OL Unit Quiz Practice

CS 7641: Machine Learning

Solutions

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1 Section 1 - Randomized Optimization

Part 1. Randomized Hill Climbing (RHC)

Correct: a, c, e **Incorrect:** b, d, f

a) At each step, RHC samples a neighbor at random and moves only if that neighbor improves the objective.
Correct. A common RHC variant samples a single neighbor uniformly at random and accepts the move only if it strictly improves the objective ("improving-only").

b) Without random restarts or sideways moves, RHC is guaranteed to eventually reach a global optimum in a finite, discrete landscape.
Incorrect. Improving-only RHC can get stuck at a *local* optimum (or plateau). Finite state space does not guarantee reaching the global optimum.

c) If you evaluate many random neighbors per step and choose the best among those samples, RHC approaches greedy hill climbing.
Correct. Sampling a large subset of neighbors and picking the best is a Monte Carlo proxy for exhaustive best-improvement (greedy) hill climbing.

d) RHC requires gradient information to choose ascent directions.
Incorrect. RHC is derivative-free; it uses only objective evaluations on neighbors.

e) On flat plateaus (all neighbors equal under a strict "improving-only" rule), RHC can stall.
Correct. With no strictly better neighbor, the algorithm has no admissible move and halts.

f) With uniform neighbor sampling, the expected improvement per step generally *increases* as you approach a local optimum.
Incorrect. Near a local optimum, improving neighbors become rarer/smaller; expected improvement tends to *decrease* toward zero.

Part 2. Simulated Annealing (SA)

Correct: a, b **Incorrect:** c, d, e, f

a) At temperature T, a worse move with cost increase $\Delta E > 0$ is accepted with probability $\exp(-\Delta E/T)$.
Correct. This is the Metropolis acceptance rule used in SA.

b) With logarithmic cooling and infinite time, SA converges to a global optimum with probability 1.
Correct. Under classical results (e.g., Geman–Geman) and suitable regularity, logarithmic cooling yields convergence in probability to the global optimum.

c) Raising T increases the penalty for uphill moves.
Incorrect. Higher T makes uphill moves *more* likely (reduces the penalty), aiding exploration.

d) SA's acceptance tests depend on absolute energies, not differences.
Incorrect. Acceptance decisions depend on the *difference* ΔE between candidate and current states.

e) SA requires gradient and Hessian information.
Incorrect. SA is derivative-free; it needs only objective evaluations.

f) Beneficial moves are accepted with probability $\exp(+\Delta E/T)$.
Incorrect. For $\Delta E < 0$ (downhill), acceptance probability is 1.

Part 3. Genetic Algorithms (Numerical)

A GA operates on fixed-length binary strings of length $l = 4$. Population size is $N = 8$. Selection is fitness-proportionate (roulette wheel) with replacement. Single-point crossover occurs with probability $p_c = 0.7$. Bitwise mutation occurs independently per bit with probability $p_m = 0.01$.

(a): Fitness-Proportionate Selection A population of 4 individuals has fitnesses [1, 2, 3, 4] with total fitness 10. Let x denote the individual with fitness 3. *Question:* What is the *expected number of copies* of x selected into the next mating pool of size $N = 8$ (selection only)?

Answer (number): 2.4

Derivation: $E[\#x] = N \cdot \frac{3}{10} = 8 \cdot 0.3 = 2.4$.

(b): Crossover Survival of a Schema Consider the schema $H = 1{*}0{*}$ (positions 1 and 3 fixed). Two parents both match H. With single-point crossover and $p_c = 0.7$, what is the probability that an offspring *still matches* H after crossover (ignoring mutation)?

Answer (number): 0.5333

Derivation: Defining length $\delta(H) = 3 - 1 = 2$, string length $l = 4 \Rightarrow l - 1 = 3$. Survival under single-point crossover:

$$P(\text{survive}) \;=\; 1 - p_c \cdot \frac{\delta(H)}{l-1} \;=\; 1 - 0.7 \cdot \frac{2}{3} \;=\; 1 - 0.466\overline{6} \;=\; 0.533\overline{3}.$$

(c): Mutation Survival of a Schema Using the same schema $H = 1{*}0{*}$ and mutation rate $p_m = 0.01$, what is the probability the schema *survives mutation* (i.e., its fixed bits are unchanged)?

Answer (number): 0.9801

Derivation: Order $o(H) = 2$ fixed positions; survival under bitwise mutation is $(1 - p_m)^{o(H)} = 0.99^2 = 0.9801$.

2 Section 2 - Deconstructing AdamW

Part 1. AdaGrad & RMSProp (MCMA)

Correct: a, b, c, d **Incorrect:** e, f

a) AdaGrad accumulates squared gradients and uses $(\sqrt{G_t}+\varepsilon)^{-1}$, shrinking steps on frequently large-gradient coordinates.
Correct. AdaGrad maintains a cumulative (elementwise) sum of past squared gradients $G_t = \sum_{i\le t} g_i^2$ and scales updates by $(\sqrt{G_t}+\varepsilon)^{-1}$, which down-weights coordinates with consistently large magnitudes.

b) AdaGrad's cumulative memory can make effective learning rates vanish late in training.
Correct. Because G_t only grows, the denominator keeps increasing; step sizes can decay toward zero, slowing or stalling late-stage progress.

c) RMSProp replaces AdaGrad's cumulative sum with an EMA of squared gradients to track nonstationary curvature.
Correct. RMSProp uses an exponential moving average of g_t^2 (e.g., $v_t = \rho v_{t-1} + (1-\rho)g_t^2$) to remain responsive when curvature/noise statistics change.

d) Typical RMSProp second-moment decay ρ lies around 0.9–0.99.
Correct. Standard practice sets ρ (a.k.a. decay) in this range, balancing smoothing and responsiveness; ε handles numerical stability.

e) AdaGrad *increases* learning rates for frequently updated features.
Incorrect. It does the opposite: frequent/large gradients inflate G_t, *reducing* the effective step size on those coordinates.

f) RMSProp requires increasing ρ during training to remain stable.
Incorrect. Stability does not require increasing ρ over time; a fixed ρ with tuned α and ε is standard.

Part 2. L2 vs. Weight Decay & AdamW (MCMA)

Correct: a, b, c, d, e **Incorrect:** f

a) Under SGD with a scalar preconditioner, adding an L2 penalty is equivalent to multiplicative weight decay each step.
Correct. For plain/scale-SGD, the penalized update $\theta \leftarrow \theta - \alpha(\nabla L + \lambda\theta)$ equals applying $(1-\alpha\lambda)$ shrinkage then a gradient step.

b) Under adaptive preconditioning, L2 shrinkage is coordinate-wise and coupled to gradient history, breaking equivalence to uniform decay.
Correct. With diagonal, data-dependent preconditioners (e.g., Adam/RMSProp), the L2 term is scaled per-coordinate, no longer matching uniform multiplicative decay.

c) Equivalence to weight decay holds iff the preconditioner P_t is a positive scalar multiple of the identity.

Correct. Only when $P_t = c_t I$ does the penalized update commute with uniform shrinkage, preserving equivalence.

d) AdamW applies decoupled shrinkage $(1 - \alpha\lambda)\theta_t$ *separately* from the adaptive step.
Correct. AdamW factors weight decay out of the adaptive update, applying it as a distinct multiplicative step on parameters.

e) In practice, tune α (step size) and λ (decay) as distinct knobs; rescale λ when steps-per-epoch change.
Correct. Decoupling lets you control optimization speed vs. parameter shrink independently; changing update frequency changes per-epoch decay, so λ should be adjusted accordingly.

f) Apply the same weight decay to biases and normalization scale/shift parameters by default.
Incorrect. Common best practice is to *exclude* biases and norm scale/shift (e.g., BatchNorm/LayerNorm) from decay to avoid harming optimization/calibration.

Part 3. Adam (Numerical)

When coding up various optimizers, it is always good to calculate updates by hand to verify our code is doing what we want. We use Adam on a scalar parameter θ with

$$\alpha = 0.1, \quad \beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \varepsilon = 10^{-8}, \quad m_0 = 0,\ v_0 = 0,\ \theta_0 = 1.0.$$

Update rules (per step t with gradient g_t):

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \\ \hat{m}_t &= \frac{m_t}{1-\beta_1^t}, \qquad \hat{v}_t = \frac{v_t}{1-\beta_2^t}, \\ \theta_{t+1} &= \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}. \end{aligned}$$

(a): First step ($g_1 = 0.5$). Compute θ_1.

$$m_1 = 0.9 \cdot 0 + 0.1 \cdot 0.5 = 0.05, \quad v_1 = 0.999 \cdot 0 + 0.001 \cdot 0.5^2 = 0.00025.$$

$$\hat{m}_1 = \frac{0.05}{1-0.9} = 0.5, \quad \hat{v}_1 = \frac{0.00025}{1-0.999} = 0.25, \quad \sqrt{\hat{v}_1} = 0.5.$$

Step $= \alpha\, \hat{m}_1 / (\sqrt{\hat{v}_1} + \varepsilon) = 0.1 \cdot 0.5 / 0.5 = 0.1$. Hence

$$\boxed{\theta_1 = 0.9000}.$$

(b): Second step ($g_2 = 0.1$). Compute θ_2 (build on Part (a)).

$$m_2 = 0.9 \cdot 0.05 + 0.1 \cdot 0.1 = 0.055, \quad v_2 = 0.999 \cdot 0.00025 + 0.001 \cdot 0.1^2 = 0.00025975.$$

$$\hat{m}_2 = \frac{0.055}{1-0.9^2} = \frac{0.055}{0.19} \approx 0.28947, \quad \hat{v}_2 = \frac{0.00025975}{1-0.999^2} = \frac{0.00025975}{0.001999} \approx 0.129905.$$

$$\sqrt{\hat{v}_2} \approx 0.36042, \quad \text{step} \approx 0.1 \cdot \frac{0.28947}{0.36042} \approx 0.08032.$$

Thus $\theta_2 = \theta_1 - \text{step} \approx 0.9000 - 0.08032 = 0.81968$.

$$\boxed{\theta_2 \approx 0.8197}.$$

(c): Third step ($g_3 = -0.2$). Compute θ_3 (build on Part (b)).

$$m_3 = 0.9 \cdot 0.055 + 0.1 \cdot (-0.2) = 0.0295, \quad v_3 = 0.999 \cdot 0.00025975 + 0.001 \cdot 0.2^2 = 0.00029949025.$$

$$\hat{m}_3 = \frac{0.0295}{1 - 0.9^3} = \frac{0.0295}{0.271} \approx 0.10886, \quad \hat{v}_3 = \frac{0.00029949025}{1 - 0.999^3} = \frac{0.00029949025}{0.002997001} \approx 0.09996.$$

$$\sqrt{\hat{v}_3} \approx 0.3162, \quad \text{step} \approx 0.1 \cdot \frac{0.10886}{0.3162} \approx 0.03443.$$

So $\theta_3 = \theta_2 - \text{step} \approx 0.81968 - 0.03443 = 0.78525$.

$$\boxed{\theta_3 \approx 0.7853}.$$

(d): Fourth step ($g_4 = 0$). Compute θ_4 (build on Part (c)).

$$m_4 = 0.9 \cdot 0.0295 + 0.1 \cdot 0 = 0.02655, \quad v_4 = 0.999 \cdot 0.00029949025 + 0 = 0.00029919076.$$

$$\hat{m}_4 = \frac{0.02655}{1 - 0.9^4} = \frac{0.02655}{0.3439} \approx 0.07726, \quad \hat{v}_4 = \frac{0.00029919076}{1 - 0.999^4} = \frac{0.00029919076}{0.003994004} \approx 0.07495.$$

$$\sqrt{\hat{v}_4} \approx 0.2737, \quad \text{step} \approx 0.1 \cdot \frac{0.07726}{0.2737} \approx 0.02823.$$

Thus $\theta_4 = \theta_3 - \text{step} \approx 0.78525 - 0.02823 = 0.75702$.

$$\boxed{\theta_4 \approx 0.7570}.$$

3 Section 3 - Information Theory

Part 1. Information Theory with a 6-Sided Die (MCMA)

You are designing a communication system where each symbol corresponds to the result of rolling a 6-sided die. Let the random variable $D \in \{1, 2, 3, 4, 5, 6\}$ represent the die outcome.

In one version, the die is **fair**. A die is **fair** if:

- $P(D = i) = \frac{1}{6}$ for all $i \in \{1, 2, 3, 4, 5, 6\}$

In another, it is **loaded** so that:

- $P(D = 6) = 0.5$
- $P(D = 1) = P(D = 2) = P(D = 3) = P(D = 4) = P(D = 5) = 0.1$

You want to analyze the **entropy and information content** of this die under both conditions.

Which of the following statements are correct? Select all that apply.
Correct Answers: A, B, C, D, E

A. The entropy of the fair die is higher than that of the loaded die.
True. The fair die has maximum entropy $H = \log_2(6) \approx 2.585$ bits. The loaded die is more predictable, thus lower entropy.

B. For the fair die, each outcome conveys the same amount of information.
True. In a uniform distribution, each outcome has the same probability and thus the same information content.

C. The outcome "6" in the loaded die has the lowest information content among all possible outcomes.
True. Since $P(D = 6) = 0.5$, it is the most expected and has the lowest information: $I = -\log_2(0.5) = 1$ bit.

D. If the die is used in a coding system, the loaded die allows for more efficient compression than the fair die.
True. Lower entropy means better average codeword length when using entropy-optimal coding schemes.

E. The maximum possible entropy for any 6-sided die occurs when all outcomes are equally likely.
True. Entropy is maximized when the distribution is uniform, as this represents maximum uncertainty.

F. If you observed 100 outcomes from the loaded die, you'd expect to extract more bits of information on average than from a fair die.
False. Each roll from the fair die carries more entropy than from the loaded die. Over 100 samples, the fair die yields more total information.

Part 2. Mutual Information and KL Divergence (MCMA)

You are analyzing the relationship between two discrete variables:

- X: Whether a user clicks on an ad (*Yes*, *No*)
- Y: Whether the ad is shown during the daytime (*Day*, *Night*)

You estimate the joint distribution $P(X, Y)$ from data and want to measure how much information Y provides about X.

You compute:

- The marginal distributions $P(X)$, $P(Y)$
- The joint distribution $P(X, Y)$
- The mutual information $I(X; Y)$
- The KL divergence $D_{\mathrm{KL}}(P(X, Y) \,\|\, P(X)P(Y))$

Which of the following statements are correct? Select all that apply.
Correct Answers: A, B, D, E, F

A. Mutual information measures how far the joint distribution $P(X, Y)$ is from the assumption of independence.
True. Mutual information is the KL divergence between the joint distribution and the product of the marginals.

B. If X and Y are independent, then the mutual information and KL divergence between $P(X, Y)$ and $P(X)P(Y)$ are both zero.
True. When X and Y are independent, the joint equals the product of marginals, so their KL divergence (and mutual information) is zero.

C. KL divergence is symmetric: $D_{\mathrm{KL}}(P \,\|\, Q) = D_{\mathrm{KL}}(Q \,\|\, P)$ for any two distributions.
False. KL divergence is not symmetric. The direction matters, making D_{KL} a pseudo-distance.

D. Mutual information is always non-negative and equals zero only when the variables are independent.
True. This follows from properties of KL divergence: it is always non-negative and zero if and only if the distributions are equal.

E. The value of mutual information depends on both the joint distribution and the marginals.
True. The mutual information formula explicitly includes both the joint $P(X, Y)$ and marginals $P(X), P(Y)$.

F. KL divergence can be used to measure how much information is lost when approximating the joint distribution with the product of marginals.
True. This interpretation is precisely what makes mutual information useful in evaluating dependence.

Part 3. Conditional Entropy and Mutual Information (Numerical)

Given (from the table).

$$P(Y = \text{Spam}) = \tfrac{40}{100} = 0.4, \quad P(Y = \text{Not}) = 0.6,$$
$$P(X = \text{Present}) = \tfrac{40}{100} = 0.4, \quad P(X = \text{Absent}) = 0.6,$$
$$P(Y = \text{Spam}, X = \text{Present}) = \tfrac{30}{100} = 0.3, \; P(Y = \text{Spam}, X = \text{Absent}) = \tfrac{10}{100} = 0.1,$$
$$P(Y = \text{Not}, X = \text{Present}) = \tfrac{10}{100} = 0.1, \; P(Y = \text{Not}, X = \text{Absent}) = \tfrac{50}{100} = 0.5.$$

All logarithms are base-2; units are bits.

a) **Marginal entropy** $H(Y)$.

$$H(Y) = -\sum_y P(y) \log_2 P(y) = -(0.4 \log_2 0.4 + 0.6 \log_2 0.6)$$

$\log_2 0.4 = -1.32193$, $\log_2 0.6 = -0.73697 \;\Rightarrow\; H(Y) = 0.4(1.32193) + 0.6(0.73697) = \boxed{0.97095 \text{ bits}}$.

b) **Conditional entropy** $H(Y \mid X)$.

$$H(Y \mid X) = \sum_x P(x)\, H(Y \mid X = x).$$

<u>Given $X = \text{Present}$:</u> $P(Y = \text{Spam} \mid X) = 30/40 = 0.75$, $P(Y = \text{Not} \mid X) = 0.25$.

$$H(Y \mid X = \text{Present}) = -(0.75 \log_2 0.75 + 0.25 \log_2 0.25) = 0.31128 + 0.5 = 0.81128.$$

<u>Given $X = \text{Absent}$:</u> $P(Y = \text{Spam} \mid X) = 10/60 = 1/6$, $P(Y = \text{Not} \mid X) = 5/6$.

$\log_2 \frac{1}{6} = -2.58496$, $\log_2 \frac{5}{6} = -0.26303 \Rightarrow H(Y \mid X = \text{Absent}) = \frac{1}{6}(2.58496) + \frac{5}{6}(0.26303) = 0.65002$.

Combine:

$$H(Y \mid X) = 0.4(0.81128) + 0.6(0.65002) = 0.32451 + 0.39001 = \boxed{0.71452 \text{ bits}}.$$

c) **Mutual information** $I(Y; X) = H(Y) - H(Y \mid X)$.

$$I(Y; X) = 0.97095 - 0.71452 = \boxed{0.25643 \text{ bits}}.$$

d) **Effect of adding 1000 more emails with the same proportions.**
The empirical *probabilities* are unchanged, so all information quantities remain the same:

$$\boxed{H(Y) = 0.97095 \text{ bits}}, \quad \boxed{H(Y \mid X) = 0.71452 \text{ bits}}, \quad \boxed{I(Y; X) = 0.25643 \text{ bits}}.$$

(With more samples the *estimation variance* would drop, but the population values above are unchanged.)

e) **Why MI is invariant to sample size but not to distribution shape.**
Mutual information depends only on the *underlying probabilities* $P(X, Y)$. Scaling the dataset size while keeping the *proportions* fixed leaves $P(X, Y)$ unchanged, so $H(Y)$, $H(Y \mid X)$, and $I(Y; X)$ are identical. Changing the *shape* (i.e., the relative frequencies/probabilities) alters $P(X, Y)$, which changes the entropies and hence $I(Y; X)$.

4 Section 4 - Bayesian Learning

Part 1. Model Comparison (MCMA)

Correct: a, c **Incorrect:** b, d, e, f

a) Posterior odds update via the likelihood ratio: $\dfrac{P(h_1 \mid D)}{P(h_2 \mid D)} = \dfrac{P(h_1)}{P(h_2)} \cdot \dfrac{P(D \mid h_1)}{P(D \mid h_2)}$.
Correct. This is the odds form of Bayes' rule: posterior odds = prior odds × Bayes factor.

b) If the likelihood ratio equals 1, the data strictly favor h_1 over h_2.
Incorrect. A likelihood ratio of 1 means the data are *equally likely* under both hypotheses; the data favor neither.

c) When comparing h_1 vs. h_2 on the same D, the evidence $P(D)$ cancels out of the odds.
Correct. Since $P(h_i \mid D) \propto P(D \mid h_i)P(h_i)$ with the same $P(D)$ in numerator/denominator, it cancels in the ratio.

d) With equal priors, $P(h \mid D) = P(D \mid h)$ for each h.
Incorrect. With equal priors, $P(h \mid D) \propto P(D \mid h)$, but equality fails because you still must normalize across h.

e) If $P(D \mid h_1) = 0$, then $P(h_1 \mid D)$ can still be positive with a strong prior.
Incorrect. The posterior $P(h_1 \mid D) \propto P(D \mid h_1)P(h_1)$ is 0 if the likelihood is 0, regardless of the prior (assuming $P(D) > 0$).

f) Bayes' rule only applies when $\{h_1, h_2\}$ list all possible hypotheses.
Incorrect. Bayes' rule applies to any set of hypotheses; exact normalization to a probability distribution requires summing over all considered hypotheses, but the rule itself is not restricted to an exhaustive set.

Part 2. Common Pitfalls (MCMA)

Correct: a **Incorrect:** b, c, d, e, f

a) Bayes' rule updates prior beliefs $P(h)$ using data through the likelihood $P(D \mid h)$ to produce $P(h \mid D)$.
Correct. Posterior $P(h \mid D) \propto P(D \mid h)P(h)$ with normalization by $P(D)$.

b) Bayes' rule states $P(D \mid h) = P(h \mid D)$.
Incorrect. Those are different conditionals; Bayes relates them via $P(h \mid D) = \dfrac{P(D \mid h)P(h)}{P(D)}$.

c) The evidence $P(D)$ is the probability of hypothesis h after observing data.
Incorrect. $P(D)$ is a *data* marginal (normalizer), not a hypothesis probability.

d) Maximum a posteriori (MAP) ignores priors and uses only the likelihood.
Incorrect. That describes ML. MAP maximizes $P(D \mid h)P(h)$, explicitly using the prior.

e) The likelihood equals the prior: $P(D \mid h) = P(h)$.
Incorrect. Likelihood is a function of data given h; the prior is a belief on h *before* seeing data.

f) If the prior on h is zero, enough data can still make $P(h \mid D) > 0$.
Incorrect. With $P(h) = 0$, the posterior remains 0 under standard Bayes (since $P(h \mid D) \propto P(D \mid h)P(h)$).

Part 3. Bayes' Rule with Ducks (Numerical)

You're birding at Lake Kaczka during spring migration. You're trying to decide whether a duck you spot is a *Mallard* (green-headed) or *Not Mallard*. Let

$$Y \in \{1, 0\} \quad \text{denote Mallard (1) vs. Not Mallard (0)},$$

and

$$X \in \{1, 0\} \quad \text{denote whether you clearly see a green head (1) or not (0) through } \textit{clear} \text{ binoculars.}$$

From past seasons at this lake:

$$P(Y{=}1) = 0.1, \qquad P(X{=}1 \mid Y{=}1) = 0.7, \qquad P(X{=}1 \mid Y{=}0) = 0.2.$$

Only in parts (d) and (e) you switch to a foggy ridgeline where the viewing is noisy. There, the *observed* bit Z is a flipped version of X with probability $\varepsilon = 0.1$:

$$P(Z{=}1 \mid X{=}1) = 0.9, \qquad P(Z{=}1 \mid X{=}0) = 0.1.$$

(a) Clear sighting (single). Compute $P(\text{Mallard} \mid X{=}1)$.

$$P(X{=}1) = 0.7 \cdot 0.1 + 0.2 \cdot 0.9 = 0.25,$$
$$P(Y{=}1 \mid X{=}1) = \frac{0.7 \cdot 0.1}{0.25} = \boxed{0.280}.$$

(b) Clear sighting (none). Compute $P(\text{Mallard} \mid X{=}0)$.

$$P(X{=}0) = 0.75, \quad P(X{=}0 \mid Y{=}1) = 0.3, \quad P(X{=}0 \mid Y{=}0) = 0.8,$$
$$P(Y{=}1 \mid X{=}0) = \frac{0.3 \cdot 0.1}{0.3 \cdot 0.1 + 0.8 \cdot 0.9} = \frac{0.03}{0.75} = \boxed{0.040}.$$

(c) Two *independent clear* sightings (both green). You and a friend both clearly see green heads: $X_1{=}X_2{=}1$ (clean optics). Compute $P(\text{Mallard} \mid X_1{=}X_2{=}1)$.

$$P(X_1{=}1, X_2{=}1 \mid Y{=}1) = 0.7^2 = 0.49,$$
$$P(X_1{=}1, X_2{=}1 \mid Y{=}0) = 0.2^2 = 0.04,$$
$$P(Y{=}1 \mid \text{both } 1) = \frac{0.49 \cdot 0.1}{0.49 \cdot 0.1 + 0.04 \cdot 0.9} = \frac{0.049}{0.049 + 0.036} = \frac{0.049}{0.085} = \boxed{0.576}.$$

(d) *Foggy* sighting (single). Now the ridgeline is foggy; you observe $Z{=}1$ with $\varepsilon = 0.1$ noise. Compute $P(\text{Mallard} \mid Z{=}1)$.

$$
\begin{aligned}
P(Z{=}1 \mid Y{=}1) &= 0.9 \cdot 0.7 + 0.1 \cdot 0.3 = 0.63 + 0.03 = 0.66, \\
P(Z{=}1 \mid Y{=}0) &= 0.9 \cdot 0.2 + 0.1 \cdot 0.8 = 0.18 + 0.08 = 0.26, \\
P(Y{=}1 \mid Z{=}1) &= \frac{0.66 \cdot 0.1}{0.66 \cdot 0.1 + 0.26 \cdot 0.9} = \frac{0.066}{0.066 + 0.234} = \frac{0.066}{0.300} = \boxed{0.220}.
\end{aligned}
$$

(e) Two *independent foggy* sightings (both green). Two observers, same duck, both report $Z_1{=}Z_2{=}1$ under the fog model. Compute $P(\text{Mallard} \mid Z_1{=}Z_2{=}1)$.

$$
\begin{aligned}
P(Z_1{=}1, Z_2{=}1 \mid Y{=}1) &= 0.66^2 = 0.4356, \\
P(Z_1{=}1, Z_2{=}1 \mid Y{=}0) &= 0.26^2 = 0.0676, \\
P(Y{=}1 \mid Z_1{=}Z_2{=}1) &= \frac{0.4356 \cdot 0.1}{0.4356 \cdot 0.1 + 0.0676 \cdot 0.9} = \frac{0.04356}{0.04356 + 0.06084} = \frac{0.04356}{0.10440} = \boxed{0.417}.
\end{aligned}
$$

5 Question 5 - Bayesian Inference

Part 1. Noisy Data (MCMA)

Correct: b, c, d **Incorrect:** a, e, f

A. Label noise in the training data always improves generalization.
Incorrect. Label noise usually *hurts* supervised learning by corrupting targets; any regularization-like effect is problem-dependent and not guaranteed.

B. Uniform label noise can degrade the performance of maximum likelihood estimation.
Correct. Standard MLE fits observed labels; with random flips it optimizes a misspecified objective, increasing expected risk unless the noise model is accounted for.

C. Noisy inputs can make decision boundaries harder to learn accurately.
Correct. Input noise blurs feature–label relationships, reducing effective signal-to-noise and shifting/softening the learned boundary.

D. Regularization can help mitigate the effects of noise.
Correct. Penalties (e.g., weight decay), early stopping, and data augmentation can reduce overfitting to noisy labels/features.

E. Noise has no impact on Bayes-optimal classifiers.
Incorrect. Noise changes the data-generating distribution; the Bayes decision rule for the *noisy* process can differ from the clean-label Bayes rule and generally yields higher irreducible error.

F. Increasing model complexity always reduces the impact of noise.
Incorrect. Higher capacity often *amplifies* overfitting to noise; bias–variance trade-offs mean more complexity is not a universal fix.

Part 2. Belief Networks (MCMA)

Correct: a, b, c **Incorrect:** d, e, f

a) The joint factorizes as $P(X_1, \ldots, X_n) = \prod_i P(X_i \mid \mathrm{Pa}(X_i))$.
Correct. This is the defining factorization of a Bayesian network with parent sets $\mathrm{Pa}(X_i)$.

b) D-separation in the DAG implies the corresponding conditional independence for any distribution that factorizes over the DAG.
Correct. D-separation provides sound (and complete for faithful models) conditional independence implications for BN-factorizing distributions.

c) Each node is independent of its non-descendants given its parents (local Markov property).
Correct. The local Markov property underpins the global factorization and many inference algorithms.

d) Directed cycles are allowed as long as all CPTs are normalized.
Incorrect. BNs require a *directed acyclic* graph; cycles violate the semantics of the factorization.

e) CPTs must be Gaussian for the BN semantics to hold.
Incorrect. CPTs can be any valid conditional distributions (categorical, Gaussian, logistic, etc.); Gaussianity is not required.

f) Moralization preserves edge directions and therefore preserves the same conditional independences.
Incorrect. Moralization *drops* directions and adds edges between co-parents to form an undirected graph for certain algorithms; it changes edge directions and can alter conditional independence structure.

Part 3. Posterior Calculation with Noisy Labels (Numerical)

In Flavortown, every dish is either a *Spicy Slammer* ($y = 1$) or a *Mellow Munch* ($y = 0$). The kitchen printer sometimes mislabels tickets:

$$P(\tilde{y} \neq y) = \eta = 0.2 \quad (20\% \text{ label flips}).$$

From past seasons, Spicy Slammers are more common:

$$P(y = 1) = 0.6, \qquad P(y = 0) = 0.4.$$

Tonight's ticket shows an observed label $\tilde{y} = 1$ (it says "Spicy Slammer"). Compute the following (round to three decimals).

a) $P(\tilde{y} = 1 \mid y = 1)$.
Because labels flip with probability η:

$$P(\tilde{y} = 1 \mid y = 1) = 1 - \eta = 1 - 0.2 = \boxed{0.800}.$$

b) $P(\tilde{y} = 1 \mid y = 0)$.
A flip from 0 to 1 happens with probability η:

$$P(\tilde{y} = 1 \mid y = 0) = \eta = 0.2 = \boxed{0.200}.$$

c) $P(y = 1 \mid \tilde{y} = 1)$ via Bayes' rule.

$$\begin{aligned} P(y = 1 \mid \tilde{y} = 1) &= \frac{P(\tilde{y} = 1 \mid y = 1)\, P(y = 1)}{P(\tilde{y} = 1 \mid y = 1)\, P(y = 1) + P(\tilde{y} = 1 \mid y = 0)\, P(y = 0)} \\ &= \frac{0.8 \cdot 0.6}{0.8 \cdot 0.6 + 0.2 \cdot 0.4} = \frac{0.48}{0.48 + 0.08} = \frac{0.48}{0.56} = \boxed{0.857}. \end{aligned}$$

d) **MAP decision.**

$$P(y = 1 \mid \tilde{y} = 1) = 0.857 \;>\; 0.143 = P(y = 0 \mid \tilde{y} = 1) \;\Rightarrow\; \text{choose } \boxed{y = 1 \text{ (Spicy Slammer)}}.$$